Lesson 3: Measurable Functions and Simple Approximation

AI/ML Mathematics

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1 Motivation and Overview

Every random variable in reinforcement learning – states, actions, rewards, returns – is a measurable function from a probability space to a target space. This lesson develops: (1) the generator criterion for checking measurability efficiently, (2) closure under algebraic operations and pointwise limits, (3) the simple function approximation theorem that bridges to Lebesgue integration, and (4) the Doob–Dynkin lemma characterizing $\sigma(X)$-measurability as "being a function of X."

2 Measurable Functions

2.1 Definition and Basic Properties

Definition 2.1 (Measurable Function)**.** Let $(\Omega, \mathcal{F})$ and $(S, \mathcal{S})$ be measurable spaces. A function $f : \Omega \to S$ is $(\mathcal{F}, \mathcal{S})$-*measurable* (or simply *measurable* when the sigma-algebras are clear) if

$$f^{-1}(B) := \{\omega \in \Omega : f(\omega) \in B\} \in \mathcal{F} \quad \text{for all } B \in \mathcal{S}. \tag{1}$$

When $(S, \mathcal{S}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ we call f *Borel measurable*, and when a probability measure $\mathbb{P}$ is present on $(\Omega, \mathcal{F})$ we call f a *random variable*.

Remark 2.2. The preimage operation satisfies $f^{-1}(B^c) = (f^{-1}(B))^c$ and $f^{-1}(\bigcup_n B_n) = \bigcup_n f^{-1}(B_n)$. These identities are used repeatedly when verifying measurability.

2.2 The Generator Criterion

Checking the preimage condition for every set in $\mathcal{S}$ is usually infeasible. The following result reduces the task to a generating collection.

Generator Criterion

Theorem 2.3 (Generator Criterion for Measurability)**.** *Let $\mathcal{S} = \sigma(\mathcal{C})$ for some collection $\mathcal{C}$ of subsets of S. Then $f : (\Omega, \mathcal{F}) \to (S, \mathcal{S})$ is measurable if and only if*

$$f^{-1}(C) \in \mathcal{F} \quad \textit{for all } C \in \mathcal{C}. \tag{2}$$

Proof. ($\Rightarrow$) If f is measurable, then $f^{-1}(B) \in \mathcal{F}$ for all $B \in \mathcal{S}$. Since $\mathcal{C} \subseteq \mathcal{S}$, the claim is immediate.

($\Leftarrow$) Define the *good-sets collection*

$$\mathcal{G} = \{B \subseteq S : f^{-1}(B) \in \mathcal{F}\}. \tag{3}$$

We show that $\mathcal{G}$ is a sigma-algebra on S.

Step 1. $S \in \mathcal{G}$: we have $f^{-1}(S) = \Omega \in \mathcal{F}$.

Step 2. If $B \in \mathcal{G}$, then $f^{-1}(B^c) = (f^{-1}(B))^c \in \mathcal{F}$, so $B^c \in \mathcal{G}$.

Step 3. If $B_1, B_2, \ldots \in \mathcal{G}$, then $f^{-1}(\bigcup_n B_n) = \bigcup_n f^{-1}(B_n) \in \mathcal{F}$ (a countable union of $\mathcal{F}$-sets), so $\bigcup_n B_n \in \mathcal{G}$.

Thus $\mathcal{G}$ is a sigma-algebra. By hypothesis, $\mathcal{C} \subseteq \mathcal{G}$. Since $\sigma(\mathcal{C})$ is the smallest sigma-algebra containing $\mathcal{C}$, we obtain $\mathcal{S} = \sigma(\mathcal{C}) \subseteq \mathcal{G}$. Hence $f^{-1}(B) \in \mathcal{F}$ for all $B \in \mathcal{S}$, i.e., f is measurable. □

Corollary 2.4 (Standard Criterion for Real-Valued Measurability)**.** *A function $f : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is measurable if and only if any one of the following equivalent conditions holds for all $a \in \mathbb{R}$:*

(i) $\{\omega : f(\omega) \leq a\} \in \mathcal{F}$,

(ii) $\{\omega : f(\omega) < a\} \in \mathcal{F}$,
(iii) $\{\omega : f(\omega) \geq a\} \in \mathcal{F}$,
(iv) $\{\omega : f(\omega) > a\} \in \mathcal{F}$.

Proof. Each of the four families generates $\mathcal{B}(\mathbb{R})$. For instance, the half-lines $\{(-\infty, a] : a \in \mathbb{R}\}$ generate $\mathcal{B}(\mathbb{R})$ (see Lesson 1), so condition (i) is equivalent to measurability by Theorem 2.3. For (ii), note that $\{f < a\} = \bigcup_{n=1}^{\infty}\{f \leq a - 1/n\}$, so (i) $\Leftrightarrow$ (ii). Conditions (iii) and (iv) follow by taking complements. □

Connection to RL

A random variable X on $(\Omega, \mathcal{F}, \mathbb{P})$ is precisely a measurable function to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Corollary 2.4 says that being a random variable means the events $\{X \leq x\}$ belong to $\mathcal{F}$, which is exactly what is needed to define the CDF $F_X(x) = \mathbb{P}(X \leq x)$. In an MDP, every quantity we compute probabilities for – S_t, A_t, R_{t+1}, the return G_t – must be a random variable in this precise sense.

Example 2.5 (Composition of Measurable Functions)**.** If $f : (\Omega, \mathcal{F}) \to (S, \mathcal{S})$ and $g : (S, \mathcal{S}) \to (T, \mathcal{T})$ are both measurable, then $g \circ f : (\Omega, \mathcal{F}) \to (T, \mathcal{T})$ is measurable. Indeed, for any $B \in \mathcal{T}$, $(g \circ f)^{-1}(B) = f^{-1}(g^{-1}(B))$. Since g is measurable, $g^{-1}(B) \in \mathcal{S}$, and since f is measurable, $f^{-1}(g^{-1}(B)) \in \mathcal{F}$. This implies: if X is a random variable and $\varphi : \mathbb{R} \to \mathbb{R}$ is Borel measurable, then $\varphi(X)$ is a random variable.

3 Closure Under Algebraic Operations

Theorem 3.1 (Closure Under Algebraic Operations)**.** *Let $f, g : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be measurable and let $c \in \mathbb{R}$. Then the following functions are measurable:*
(a) $cf, \quad f + g, \quad fg, \quad |f|$,
(b) $\max(f, g)$ *and* $\min(f, g)$,
(c) f/g *on* $\{\omega : g(\omega) \neq 0\}$,
(d) $f^+ = \max(f, 0)$ *and* $f^- = \max(-f, 0)$.

Proof. We give complete proofs for the key cases.

Scalar multiple cf. If $c = 0$, cf is constant and hence measurable. If $c > 0$, $\{cf \leq a\} = \{f \leq a/c\} \in \mathcal{F}$. If $c < 0$, $\{cf \leq a\} = \{f \geq a/c\} \in \mathcal{F}$ by Corollary 2.4(iii).

Sum $f + g$. Fix $a \in \mathbb{R}$. We claim

$$\{f + g < a\} = \bigcup_{q \in \mathbb{Q}} \big(\{f < q\} \cap \{g < a - q\}\big). \tag{4}$$

Proof of (4). If $f(\omega) + g(\omega) < a$, set $\varepsilon = a - f(\omega) - g(\omega) > 0$. By density of $\mathbb{Q}$ in $\mathbb{R}$, there exists $q \in \mathbb{Q}$ with $f(\omega) < q < f(\omega) + \varepsilon = a - g(\omega)$. Then $g(\omega) < a - q$, so ω belongs to the right-hand side. Conversely, if $f(\omega) < q$ and $g(\omega) < a - q$ for some q, then $f(\omega) + g(\omega) < a$.

Each set $\{f < q\}$ and $\{g < a - q\}$ is in $\mathcal{F}$. The union over the countable set $\mathbb{Q}$ yields $\{f + g < a\} \in \mathcal{F}$. Since $\{f + g \leq a\} = \bigcap_{n=1}^{\infty}\{f + g < a + 1/n\}$ is a countable intersection of $\mathcal{F}$-sets, we get measurability of $f + g$.

Product fg. First observe that f^2 is measurable: for $a < 0$ we have $\{f^2 \leq a\} = \emptyset \in \mathcal{F}$, and for $a \geq 0$, $\{f^2 \leq a\} = \{-\sqrt{a} \leq f \leq \sqrt{a}\} = \{f \leq \sqrt{a}\} \cap \{f \geq -\sqrt{a}\} \in \mathcal{F}$. Now use the polarization identity $fg = \frac{1}{4}\big[(f + g)^2 - (f - g)^2\big]$, which expresses the product in terms of sums and squarings.

Absolute value $|f|$. We have $\{|f| \leq a\} = \{-a \leq f \leq a\} \in \mathcal{F}$ for $a \geq 0$ and $\{|f| \leq a\} = \emptyset$ for $a < 0$.

Maximum and minimum. $\{\max(f,g) \le a\} = \{f \le a\} \cap \{g \le a\} \in \mathcal{F}$. $\min(f,g) = -\max(-f,-g)$ is then measurable by the scalar-multiple case.

Quotient f/g. On $\{g \neq 0\}$ the function $1/g$ is measurable (by a direct check using the standard criterion), and $f/g = f \cdot (1/g)$ is measurable by the product case.

Positive and negative parts. $f^+ = \max(f,0)$ and $f^- = \max(-f,0)$ are covered by the maximum case with $g = 0$ (constant, hence measurable). □

Remark 3.2. Note that $f = f^+ - f^-$ and $|f| = f^+ + f^-$. These decompositions are used when defining the Lebesgue integral: one first integrates non-negative functions, then defines $\int f = \int f^+ - \int f^-$.

4 Limits of Measurable Functions

Theorem 4.1 (Closure Under Pointwise Limits)**.** *Let $(f_n)_{n\ge1}$ be a sequence of measurable functions from $(\Omega,\mathcal{F})$ to $(\mathbb{R},\mathcal{B}(\mathbb{R}))$. Then the following extended-real-valued functions are measurable:*

$$\sup_n f_n, \quad \inf_n f_n, \quad \limsup_{n\to\infty} f_n, \quad \liminf_{n\to\infty} f_n. \tag{5}$$

In particular, if $f(\omega) = \lim_{n\to\infty} f_n(\omega)$ exists for all ω (or almost all ω), then f is measurable.

Proof. **Supremum.** For any $a \in \mathbb{R}$,

$$\{\sup_n f_n \le a\} = \bigcap_{n=1}^{\infty} \{f_n \le a\}. \tag{6}$$

Proof of (6). $\sup_n f_n(\omega) \le a$ if and only if $f_n(\omega) \le a$ for every n, which is $\omega \in \bigcap_n \{f_n \le a\}$.

Since each $\{f_n \le a\} \in \mathcal{F}$ and sigma-algebras are closed under countable intersection, $\{\sup_n f_n \le a\} \in \mathcal{F}$.

Infimum. Observe that $\inf_n f_n = -\sup_n(-f_n)$. Since $-f_n$ is measurable (Theorem 3.1(a) with $c = -1$), the supremum result applies.

Limsup and liminf. By definition,

$$\limsup_n f_n = \inf_{n\ge1} \sup_{k\ge n} f_k, \qquad \liminf_n f_n = \sup_{n\ge1} \inf_{k\ge n} f_k. \tag{7}$$

For each n, $g_n = \sup_{k\ge n} f_k$ is measurable by the supremum result. Then $\limsup_n f_n = \inf_n g_n$ is measurable by the infimum result. The argument for liminf is symmetric.

Pointwise limit. If $f_n \to f$ pointwise, then $f = \limsup_n f_n = \liminf_n f_n$, hence f is measurable. □

Remark 4.2. The result requires *countable* suprema and infima. An uncountable supremum of measurable functions need not be measurable.

5 Simple Functions and Approximation

5.1 Simple Functions

Definition 5.1 (Simple Function)**.** A *simple function* is a measurable function $\varphi : (\Omega,\mathcal{F}) \to \mathbb{R}$ that takes only finitely many values. Every simple function admits the *standard (canonical) representation*

$$\varphi = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i}, \tag{8}$$

where $a_1,\dots,a_n \in \mathbb{R}$ are the *distinct* values of φ, $A_i = \varphi^{-1}(\{a_i\}) \in \mathcal{F}$, and $\{A_1,\dots,A_n\}$ is a partition of Ω.

Remark 5.2. A simple function can also be written as $\sum_{j=1}^{m} b_j \mathbf{1}_{B_j}$ with $B_j \in \mathcal{F}$ (not necessarily disjoint, values not necessarily distinct). The standard representation is unique and is used when defining the Lebesgue integral of simple functions.

Proposition 5.3 (Algebra of Simple Functions)**.** *If φ and ψ are simple and $c \in \mathbb{R}$, then $c\varphi$, $\varphi + \psi$, $\varphi\psi$, $\max(\varphi, \psi)$, and $\min(\varphi, \psi)$ are all simple.*

Proof. Each operation produces a function taking finitely many values (at most the product of the number of values of φ and ψ). Measurability follows from Theorem 3.1. □

5.2 The Simple Function Approximation Theorem

Simple Function Approximation

Theorem 5.4 (Simple Function Approximation Theorem)**.** *Let $f : (\Omega, \mathcal{F}) \to [0, \infty]$ be measurable. Then there exists a sequence of simple functions $0 \le \varphi_1 \le \varphi_2 \le \cdots$ such that $\varphi_n(\omega) \uparrow f(\omega)$ for every $\omega \in \Omega$. If f is bounded, the convergence is uniform.*
For a general measurable $f : \Omega \to [-\infty, \infty]$, there exist simple functions (ψ_n) with $|\psi_n| \le |f|$ and $\psi_n \to f$ pointwise.

Proof. **Part 1: Non-negative case.** For each integer $n \ge 1$, define the simple function φ_n by a "truncation + discretization" procedure. The range $[0, \infty)$ is partitioned into $n \cdot 2^n$ intervals of width $1/2^n$ from 0 to n, plus the overflow region $[n, \infty]$:

$$\varphi_n(\omega) = \begin{cases} \dfrac{k-1}{2^n} & \text{if } \frac{k-1}{2^n} \le f(\omega) < \frac{k}{2^n}, \quad k = 1, 2, \ldots, n \cdot 2^n, \\ n & \text{if } f(\omega) \ge n. \end{cases} \tag{9}$$

In words, φ_n rounds f down to the nearest multiple of $1/2^n$, capped at n.

Step 1: φ_n is simple and measurable. The function φ_n takes at most $n \cdot 2^n + 1$ values. The level sets are

$$\varphi_n^{-1}\left(\left\{\frac{k-1}{2^n}\right\}\right) = \left\{\frac{k-1}{2^n} \le f < \frac{k}{2^n}\right\} = \left\{f \ge \frac{k-1}{2^n}\right\} \cap \left\{f < \frac{k}{2^n}\right\} \in \mathcal{F},$$

and $\varphi_n^{-1}(\{n\}) = \{f \ge n\} \in \mathcal{F}$.

Step 2: $\varphi_n \le \varphi_{n+1}$ (monotonicity). The grid at level $n+1$ is finer: each interval $[(k-1)/2^n, k/2^n)$ is split into $[(2k-2)/2^{n+1}, (2k-1)/2^{n+1})$ and $[(2k-1)/2^{n+1}, 2k/2^{n+1})$. Since φ_{n+1} rounds down on a finer grid, it gives a value at least as large as φ_n on each sub-interval. Moreover, the cap increases from n to $n+1$.

Step 3: $\varphi_n \uparrow f$ (convergence). Fix $\omega \in \Omega$.

Case (a): $f(\omega) < \infty$. Choose $N > f(\omega)$. For all $n \ge N$, we have $f(\omega) < n$ (so the cap is not hit), and $\varphi_n(\omega)$ is obtained by rounding $f(\omega)$ down to a multiple of $1/2^n$. Therefore

$$0 \le f(\omega) - \varphi_n(\omega) < \frac{1}{2^n} \to 0 \quad \text{as } n \to \infty.$$

Case (b): $f(\omega) = \infty$. For every n, $\varphi_n(\omega) = n$, so $\varphi_n(\omega) \to \infty = f(\omega)$.

Step 4: Uniform convergence when f is bounded. If $f \le M$ for some $M < \infty$, then for all $n > M$ and every ω, the cap is never reached, so $|f(\omega) - \varphi_n(\omega)| < 1/2^n$. Hence $\|f - \varphi_n\|_\infty < 1/2^n \to 0$.

Part 2: General case. Write $f = f^+ - f^-$ where $f^+ = \max(f, 0) \ge 0$ and $f^- = \max(-f, 0) \ge 0$ are both non-negative measurable functions. By Part 1, there exist increasing simple sequences $\varphi_n \uparrow f^+$ and $\chi_n \uparrow f^-$. Set $\psi_n = \varphi_n - \chi_n$. Then ψ_n is simple, $|\psi_n| = \varphi_n + \chi_n \le f^+ + f^- = |f|$, and $\psi_n \to f^+ - f^- = f$ pointwise. □

Connection to RL

The approximation theorem is the foundation of the Lebesgue integral. The integral of a non-negative function is defined as $\int f\,d\mu = \sup\{\int \varphi\,d\mu : \varphi \text{ simple}, 0 \le \varphi \le f\}$, where simple-function integrals are weighted sums. In RL, the expectation $\mathbb{E}[G_t \mid S_t = s]$ is the Lebesgue integral of the return G_t against the conditional probability measure. The approximation theorem guarantees that this integral can always be computed as a limit of sums – finite-state MDPs effectively use simple function approximations.

6 The σ-Algebra Generated by a Function

Definition 6.1 ($\sigma(X)$)**.** Let $X : (\Omega, \mathcal{F}) \to (S, \mathcal{S})$ be a measurable function. The *σ-algebra generated by X* is

$$\sigma(X) = \{X^{-1}(B) : B \in \mathcal{S}\} = \big\{\{\omega : X(\omega) \in B\} : B \in \mathcal{S}\big\}. \tag{10}$$

More generally, for a family of functions $(X_i)_{i\in I}$, $\sigma\big((X_i)_{i\in I}\big) = \sigma\big(\bigcup_{i\in I} \sigma(X_i)\big)$.

Proposition 6.2. *$\sigma(X)$ is the smallest sub-σ-algebra of $\mathcal{F}$ making X measurable.*

Proof. First, $\sigma(X) \subseteq \mathcal{F}$ because X is $(\mathcal{F}, \mathcal{S})$-measurable and therefore $X^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{S}$.

Second, X is $\sigma(X)$-measurable by construction: for any $B \in \mathcal{S}$, $X^{-1}(B) \in \sigma(X)$ by definition.

Third, if $\mathcal{G} \subseteq \mathcal{F}$ is any sub-σ-algebra with X being $\mathcal{G}$-measurable, then $X^{-1}(B) \in \mathcal{G}$ for all $B \in \mathcal{S}$, so $\sigma(X) \subseteq \mathcal{G}$. □

Example 6.3. Let $A \in \mathcal{F}$ and $X = \mathbf{1}_A$. Then $\sigma(X) = \{\emptyset, A, A^c, \Omega\}$. Indeed, $X^{-1}(\{1\}) = A$, $X^{-1}(\{0\}) = A^c$, $X^{-1}(\{0,1\}) = \Omega$, $X^{-1}(\emptyset) = \emptyset$, and these four sets form a sigma-algebra.

7 The Doob–Dynkin Lemma

The following result characterizes when a function Y is $\sigma(X)$-measurable: it must be a Borel function of X.

Doob–Dynkin Lemma

Theorem 7.1 (Doob–Dynkin Lemma)**.** *Let $X : \Omega \to S$ be a function (where $(S, \mathcal{S})$ is a measurable space) and let $Y : \Omega \to \mathbb{R}$. Then Y is $\sigma(X)$-measurable if and only if there exists a measurable function $g : (S, \mathcal{S}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that $Y = g \circ X$.*

Proof. ($\Leftarrow$) Suppose $Y = g(X)$ for some measurable $g : S \to \mathbb{R}$. For any $B \in \mathcal{B}(\mathbb{R})$,

$$Y^{-1}(B) = (g \circ X)^{-1}(B) = X^{-1}(g^{-1}(B)).$$

Since g is measurable, $g^{-1}(B) \in \mathcal{S}$, so $X^{-1}(g^{-1}(B)) \in \sigma(X)$ by definition of $\sigma(X)$. Hence Y is $\sigma(X)$-measurable.

($\Rightarrow$) We prove that if Y is $\sigma(X)$-measurable, then $Y = g(X)$ for some measurable g. The proof proceeds in three steps.

Step 1: Indicator functions. Suppose $Y = \mathbf{1}_A$ where $A \in \sigma(X)$. By definition of $\sigma(X)$, there exists $B \in \mathcal{S}$ such that $A = X^{-1}(B)$. Define $g = \mathbf{1}_B : S \to \mathbb{R}$. Then for any $\omega \in \Omega$:

$$g(X(\omega)) = \mathbf{1}_B(X(\omega)) = \begin{cases} 1 & \text{if } X(\omega) \in B, \\ 0 & \text{if } X(\omega) \notin B, \end{cases} = \mathbf{1}_{X^{-1}(B)}(\omega) = \mathbf{1}_A(\omega) = Y(\omega).$$

Moreover, $g = \mathbf{1}_B$ is $(\mathcal{S}, \mathcal{B}(\mathbb{R}))$-measurable because $g^{-1}(\{1\}) = B \in \mathcal{S}$.

Step 2: Simple functions. Suppose $Y = \sum_{i=1}^n a_i \mathbf{1}_{A_i}$ is simple with each $A_i \in \sigma(X)$. By Step 1, for each i there exists $B_i \in \mathcal{S}$ with $A_i = X^{-1}(B_i)$ and $\mathbf{1}_{A_i} = \mathbf{1}_{B_i} \circ X$. Define $g = \sum_{i=1}^n a_i \mathbf{1}_{B_i}$. Then g is measurable (a finite linear combination of measurable functions) and

$$g(X(\omega)) = \sum_{i=1}^n a_i \mathbf{1}_{B_i}(X(\omega)) = \sum_{i=1}^n a_i \mathbf{1}_{A_i}(\omega) = Y(\omega).$$

Step 3: General non-negative, then general. If $Y \geq 0$ is $\sigma(X)$-measurable, the approximation theorem (Theorem 5.4) provides simple functions $\varphi_n \uparrow Y$ with each φ_n being $\sigma(X)$-measurable (since the construction uses only level sets of Y, which are in $\sigma(X)$). By Step 2, $\varphi_n = g_n \circ X$ for measurable $g_n : S \to \mathbb{R}$.

Define $g(s) = \lim_{n\to\infty} g_n(s)$ wherever this limit exists, and $g(s) = 0$ otherwise. Since the g_n are measurable and pointwise limits of measurable functions are measurable (Theorem 4.1), g is measurable. For any $\omega \in \Omega$:

$$g(X(\omega)) = \lim_n g_n(X(\omega)) = \lim_n \varphi_n(\omega) = Y(\omega).$$

For general Y, write $Y = Y^+ - Y^-$. Each part is non-negative and $\sigma(X)$-measurable, so $Y^+ = g_+(X)$ and $Y^- = g_-(X)$, giving $Y = (g_+ - g_-)(X)$ with $g = g_+ - g_-$ measurable. □

Connection to RL

The Doob–Dynkin lemma rigorously justifies two fundamental operations in RL theory:
1. The pull-out property. In conditional expectation, $\mathbb{E}[h(X) \cdot Y \mid X] = h(X) \cdot \mathbb{E}[Y \mid X]$. Since $h(X)$ is $\sigma(X)$-measurable (it is a function of X), it behaves as a "known constant" given X and can be pulled out.
2. Value functions as functions of state. $V^\pi(S_t)$ is $\sigma(S_t)$-measurable because it is a Borel function V^π applied to S_t. When we write $\mathbb{E}[\delta_t \mid S_t]$ in TD proofs and pull out $V(S_t)$, the Doob–Dynkin lemma is the underlying justification.

8 Exercises

Exercise 8.1. Prove that every continuous function $f : \mathbb{R} \to \mathbb{R}$ is Borel measurable. *Hint:* Show that for every open set $U \subseteq \mathbb{R}$, $f^{-1}(U)$ is open, and recall that every open set is a Borel set. Then apply the generator criterion with the open sets as generators.

Exercise 8.2. Prove that every monotone function $f : \mathbb{R} \to \mathbb{R}$ is Borel measurable. *Hint:* For f non-decreasing, show that $\{f \leq a\}$ is either $\emptyset$, $\mathbb{R}$, $(-\infty, c]$, or $(-\infty, c)$ for some c, all of which are Borel sets.

Exercise 8.3. Let $f, g : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be measurable. Prove that $\{f < g\}$, $\{f \leq g\}$, and $\{f = g\}$ are in $\mathcal{F}$. *Hint:* $\{f < g\} = \bigcup_{q\in\mathbb{Q}}(\{f < q\} \cap \{q < g\})$.

Exercise 8.4. Let $A, B \in \mathcal{F}$. Express $\mathbf{1}_{A\cup B}$, $\mathbf{1}_{A\cap B}$, and $\mathbf{1}_{A^c}$ in terms of $\mathbf{1}_A$ and $\mathbf{1}_B$ using algebraic operations. Verify measurability using Theorem 3.1.

Exercise 8.5. Let $\Omega = \{1, 2, 3, 4, 5, 6\}$ and $X(\omega) = \omega \bmod 2$ (so X maps odd numbers to 1 and even to 0). List all elements of $\sigma(X)$. Then determine which of the following functions are $\sigma(X)$-measurable and, for those that are, find the Borel function g such that $Y = g(X)$:
(a) $Y(\omega) = (-1)^\omega$,
(b) $Y(\omega) = \omega^2$,
(c) $Y(\omega) = 3 \cdot \mathbf{1}_{\{1,3,5\}}(\omega) + 7 \cdot \mathbf{1}_{\{2,4,6\}}(\omega)$.

Exercise 8.6 (RL Application)**.** Consider a finite MDP with states $\mathcal{S} = \{s_1, s_2, s_3\}$ and the random variable S_t denoting the state at time t.

(a) Let $V : \mathcal{S} \to \mathbb{R}$ be a value function. Use the Doob–Dynkin lemma to show that $V(S_t)$ is $\sigma(S_t)$-measurable.

(b) The TD error is $\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$. Show that δ_t is $\sigma(S_t, A_t, R_{t+1}, S_{t+1})$-measurable.

(c) Explain why $V(S_t)$ can be pulled out of $\mathbb{E}[\delta_t \mid \mathcal{F}_t]$ where $\mathcal{F}_t = \sigma(S_0, A_0, \ldots, S_t)$.

9 Summary

Result	**Statement**	**Role**
Measurable function	$f^{-1}(B) \in \mathcal{F}$ for all $B \in \mathcal{S}$	Defines random variables
Generator criterion	Check preimages on generators only	Practical measurability test
Algebraic closure	Sums, products, absolute values of measurable functions are measurable	Build new RVs from old
Limit closure	sup, inf, $\limsup$, $\liminf$ of measurable sequences are measurable	Pointwise limits OK
Approximation theorem	$f \geq 0$ measurable $\Rightarrow \varphi_n$ simple with $\varphi_n \uparrow f$	Foundation of Lebesgue integral
Doob–Dynkin	Y is $\sigma(X)$-measurable iff $Y = g(X)$	Pull-out property

Next lesson: From Measures to Probability Spaces – we specialize measure theory to the probabilistic setting, defining probability spaces, random variables as measurable functions, distributions, and CDFs.

References